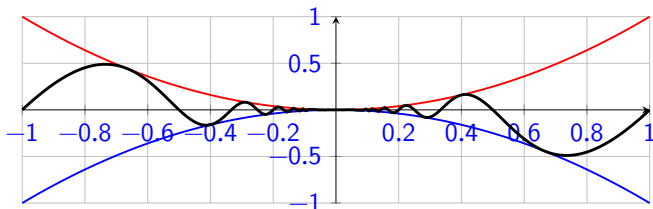


Limits

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We'll discuss the following in a rational manner:

- 1 Recall... Limits
 - Properties of Limits
 - Rational functions
- 2 Limits of Rational Functions
- 3 Completing the square
- 4 Exercises

For more, see Chapter 2 (Limits) of Strang and Herman's **Calculus**, especially Section and 2.3 (Limit Laws).

Slides are on canvas, and at niallmadden.ie/2627-MA140



Recall... Limits

Yesterday, we learned that

$$\lim_{x \rightarrow a} f(x) = L,$$

means that we can make $f(x)$ as close to L as we like, by taking x as close to a as needed.

We learned about the following “Limit Laws”: *Suppose that*

$$\lim_{x \rightarrow a} f_1(x) = L_1 \quad \text{and} \quad \lim_{x \rightarrow a} f_2(x) = L_2,$$

and $c \in \mathbb{R}$ *is any constant.* Then,

$$(1) \quad \lim_{x \rightarrow a} c = c, \quad c \in \mathbb{R}$$

$$(2) \quad \lim_{x \rightarrow a} x = a$$

$$(3) \quad \lim_{x \rightarrow a} [c f_1(x)] = c L_1$$

$$(4) \quad \lim_{x \rightarrow a} [f_1(x) + f_2(x)] = L_1 + L_2$$

and

$$\lim_{x \rightarrow a} [f_1(x) - f_2(x)] = L_1 - L_2$$

$$(5) \quad \lim_{x \rightarrow a} (f_1(x) f_2(x)) = L_1 L_2$$

$$(6) \quad \lim_{x \rightarrow a} ((f_1(x))^n) = (L_1)^n$$

$$(7) \quad \lim_{x \rightarrow a} \left(\frac{f_1(x)}{f_2(x)} \right) = \frac{L_1}{L_2},$$

providing $L_2 \neq 0$.

$$(8) \quad \lim_{x \rightarrow a} \sqrt[n]{f_1(x)} = \sqrt[n]{L_1}$$

In many cases, evaluating limits is more complicated. In particular, we'll consider numerous examples where we want to evaluate $\lim_{x \rightarrow a} f(x)$ where a is not in the domain of f .

A typical example of this is when we evaluate a rational function, $\lim_{x \rightarrow a} \frac{p(x)}{q(x)}$ where **both** $p(a) = 0$ and $q(a) = 0$.

Since we care about the value of p and q **near** $x = a$, but not actually at $x = a$, we said it is safe to factor out an $(x - a)$ term from both.

We finished with showing that

$$(a) \lim_{x \rightarrow 0} \frac{x}{x} = 1 \quad (b) \lim_{x \rightarrow 0} \frac{x^2}{x} = 0$$

$$(c) \lim_{x \rightarrow 0} \frac{x}{x^2} \text{ is not defined}$$

Limits of Rational Functions

When trying to evaluate $\lim_{x \rightarrow a} \frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are both polynomials.

- ▶ Check that $\frac{p(a)}{q(a)} = \frac{0}{0}$, which is called the **Indeterminate Form**.
- ▶ If so, we should be able to factor $(x - a)$ from both.
- ▶ Do that, and so simplify $\frac{p(x)}{q(x)}$ (for $x \neq 0$, of course).
- ▶ Try to evaluate the limit of the resulting simpler rational function.

Limits of Rational Functions

Example

Evaluate the limit

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}$$

Limits of Rational Functions

In that last example, we found that

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{x + 2}{x}$$

But these are different functions:

Limits of Rational Functions

Evaluate the limit

$$\lim_{x \rightarrow 2} \left(\frac{\frac{1}{2} - \frac{1}{x}}{x - 2} \right)$$

Completing the square

Very often, we'll evaluate limits of the form:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

where f and g are not polynomials. Some of the same ideas still apply.

Example

Evaluate

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - 1}{x^2}$$

Completing the square

Exercise 2.3.1

Evaluate the following limits

$$(a) \lim_{x \rightarrow \frac{1}{2}} \frac{x - \frac{1}{2}}{x^2 - \frac{1}{4}}$$

$$(b) \lim_{x \rightarrow -4} \frac{x^2 + 3x - 4}{x^2 + x - 12}$$

Exercise 2.3.2

(From 2023/2024 MA140 exam, Q1(a)) Evaluate the limit

$$\lim_{x \rightarrow 4} \frac{x - 4}{(\sqrt{x} - 2)(x + 9)}$$